

Evidence of a hardening in the cosmic ray proton spectrum at around 166 TeV observed by the GRAPES-3 experiment - Supplemental Material

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S1. MONTE CARLO (MC) SIMULATION DATASETS

The MC datasets for five primary groups (proton (H), helium (He), nitrogen (N), aluminium (Al) and iron (Fe)) were simulated using the CORSIKA package [1] (version 7.6900) with QGSJET-II-04 [2] and FLUKA [3] hadronic interaction models above and below 80 GeV, respectively. The elements N, Al, and Fe represent respectively the average behaviour of light (C, N, O), medium (Mg, Al, Si), and heavy (Mn, Fe, Co) mass groups in the primary cosmic rays (PCRs). The process of determining the mass composition utilizes the muon multiplicity distributions of the observed data and simulated elements. The muon multiplicity distribution of neighboring elements (beyond He) exhibits significant overlap. Therefore, simulation of individual elements at higher mass would not provide a good resolution. Hence, only five aforementioned primaries are simulated in this analysis. The datasets were simulated in an energy range from 1 TeV to 10 PeV and zenith angle range from 0° to 45° with 6.1×10^7 extensive air showers (EASs) for each primary, by assuming a spectral index of -2.5 . For a given primary, each EAS was randomly thrown ten times in a circular area of radius 150 m from the center of the scintillator detector (SD) array, named set-1. Another reconstructed dataset (set-2) was generated for the energy > 100 TeV by randomly throwing each EAS ten times in a circular area of radius 60 m from the center of the array to improve the statistics at higher energy.

Datasets with spectral index -2.7 and proposed by Gaisser-Stanev-Tilav (GST) [4] as well as H4a [5] composition models are derived from the pre-generated data with spectral index -2.5 , using a random number generator and by assigning an appropriate statistical weight factor. The dataset with spectral index -2.7 was used for the main analysis, and datasets with spectral features proposed by the GST and H4a models were used to estimate various systematic and quality checks for the unfolding procedure.

S2. MUON MULTIPLICITY DISTRIBUTION

A. Modeling of MMDs

For this analysis, the observed as well as simulated data in a shower size (N_e) range from $10^{4.0}$ to $10^{6.0}$ was used with an equal bin-width of 0.2 on a logarithmic scale. For each N_e bin of interest, the normalized muon multiplicity distribution (MMD) of each simulated

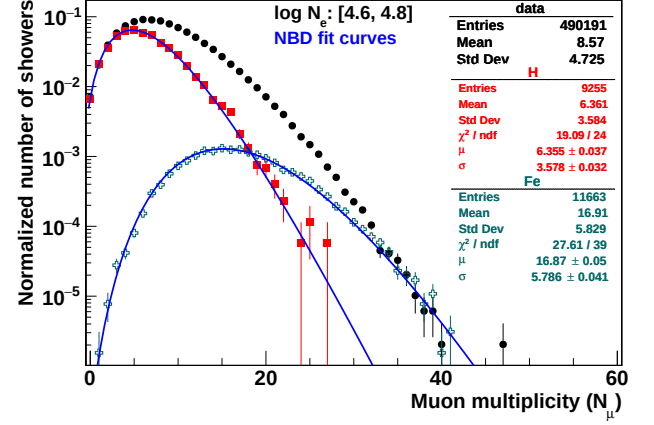


FIG. S1. Normalized MMD of the observed data (black circle) plotted along with MMDs of simulated primaries H (red square) and Fe (hollow cyan plus) for $4.6 \leq \log(N_e) < 4.8$. The blue curves represent the NBD fits using Eq. S1. The MMDs of H and Fe, along with corresponding fit curves, are scaled to overlap with the tails of the observed MMD.

primary was fitted with a negative binomial distribution (NBD) function to model the statistical uncertainties. The NBD function is given as,

$$NBD(N_\mu; \mu, \sigma) = C \cdot \left(\frac{\mu}{\sigma^2} \right)^{\frac{\mu^2}{\sigma^2 - \mu}} \left(\frac{\sigma^2 - \mu}{\sigma^2} \right)^{N_\mu},$$

$$C = \frac{\Gamma(N_\mu + \frac{\mu^2}{\sigma^2 - \mu})}{\Gamma(N_\mu + 1) \Gamma(\frac{\mu^2}{\sigma^2 - \mu})}, \quad (\text{S1})$$

where μ and σ are fit parameters representing the mean and standard deviation of the MMD, respectively, N_μ is the muon multiplicity value and Γ represents the gamma function. The MMDs of H and Fe, along with the fit curves, are plotted against the normalized observed data MMD in FIG. S1 for $4.6 \leq \log(N_e) < 4.8$. The mean and standard deviation of MMDs and fit parameters (μ and σ) are also mentioned. The μ and σ exhibit a good agreement with the mean and standard deviation of the corresponding MMD. The MMDs of H and Fe (and corresponding fit curves) are scaled with 0.53 and 0.02, respectively, so that the lower multiplicity of the observed MMD overlaps with the MMD of H and higher multiplicity of the observed MMD overlaps with Fe. At low multiplicity, the observed MMD is well described by the H primaries, while the Fe primaries well describe higher multiplicity. However, primaries of intermediate-mass numbers are required to describe the middle range of

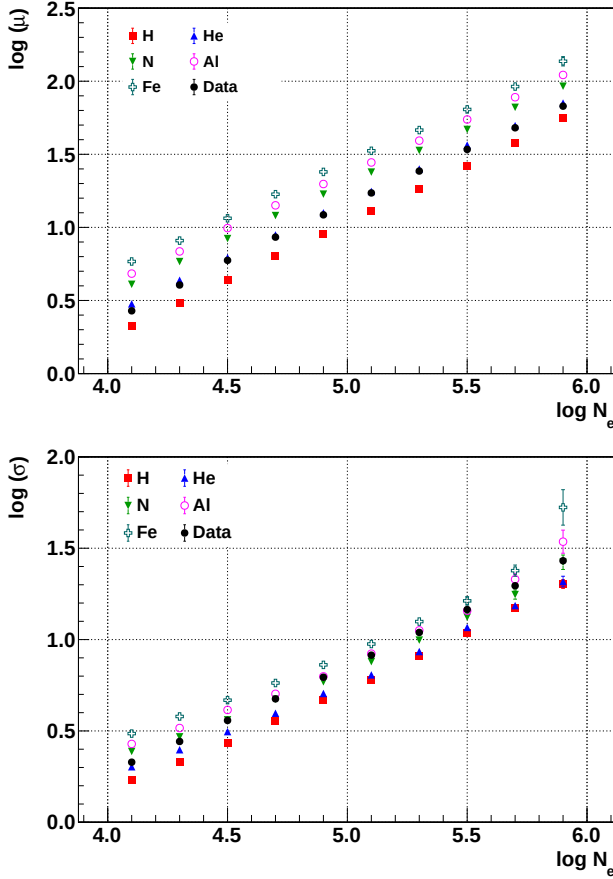


FIG. S2. The μ (top) and σ (bottom) obtained by fitting the NBD function to simulated MMDs, as a function of shower size on a log-log scale. The statistical mean and standard deviation of the observed data MMD are also plotted.

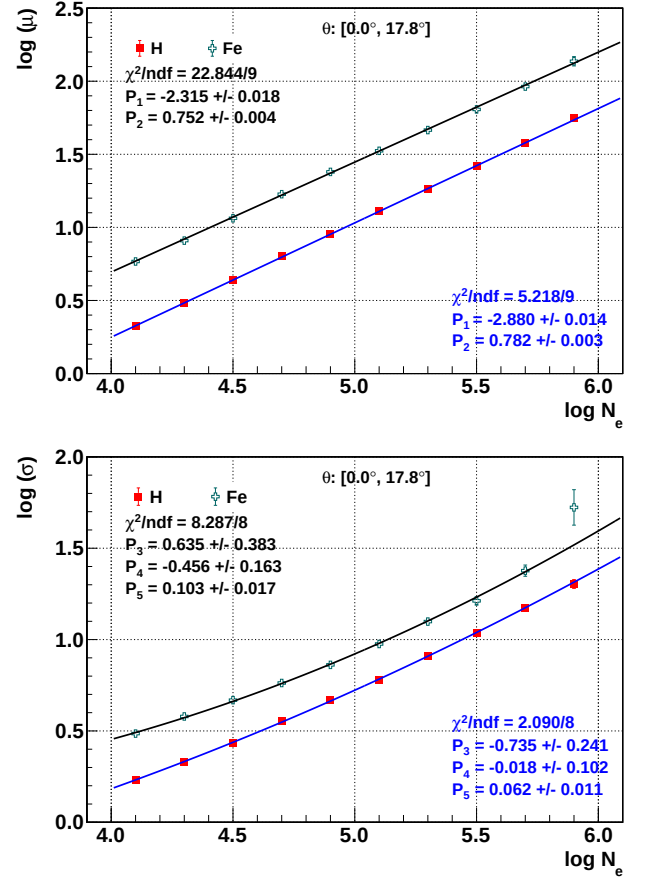


FIG. S3. The plot showing the variation of the μ (top) and σ (bottom) with shower size for H and Fe primaries, along with the parameterization curves represented by solid curves. The χ^2/ndf and fit parameters are mentioned.

the observed data MMD. The top and bottom panels of FIG. S2 show the variation of fitted μ and σ with N_e for all simulated primaries on a log-log scale. The mean and standard deviation of observed MMD are also plotted. We find that these are confined between the corresponding fit parameters of H and Fe.

B. Parameterization of fitted μ and σ

For each simulated primary, the fitted μ and σ were further parameterized as a function of N_e . The variation of μ and σ with N_e was parameterized with a linear and a second-order polynomial function, respectively, as given below,

$$\begin{aligned} \log(\mu) &= P_1 + P_2 \log N_e, \\ \log(\sigma) &= P_3 + P_4 \log N_e + P_5 (\log N_e)^2, \end{aligned} \quad (S2)$$

where P_1, P_2, P_3, P_4 and P_5 are the fit parameters. The parameterization of μ and σ for H and Fe primaries are shown in the top and bottom panels of FIG. S3, and the

TABLE S1. Fit parameters obtained from the parameterization of the μ and σ with shower size (N_e) for each simulated primary.

Sim.	Parameters for μ		Parameters for σ		
primary	P_1	P_2	P_3	P_4	P_5
H	-2.880	0.782	-0.735	-0.018	0.062
He	-2.655	0.766	-0.262	-0.161	0.073
N	-2.490	0.758	0.356	-0.383	0.095
Al	-2.404	0.755	0.764	-0.539	0.112
Fe	-2.315	0.752	0.635	-0.456	0.103

fit parameters for all simulated primaries are listed in TABLE S1.

A comparison of the NBD curve obtained by fitting the MMD with Eq. S1 (represented by the black dashed curve) and the NBD curve obtained using the μ and σ estimated from the parameterization (represented by the blue curve) for H and Fe primaries is shown in FIG. S4 for $4.6 \leq \log(N_e) < 4.8$. Both curves show very good agreement with each other for both primaries.

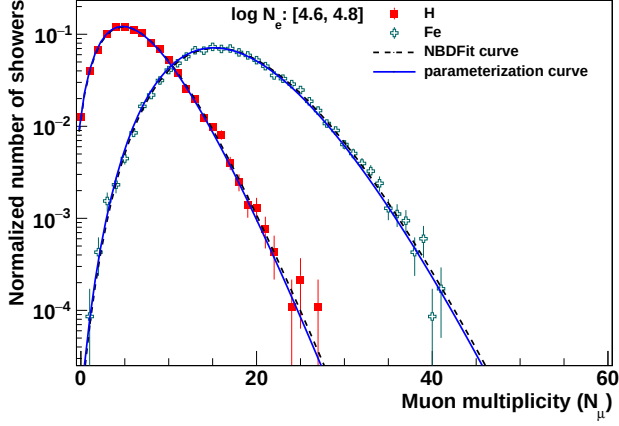


FIG. S4. A comparison of the NBD curve obtained by fitting MMD with the NBD function (black dashed curve) with the NBD curve obtained using the μ and σ estimated from the parameterization (blue curve) for H and Fe primaries and $4.6 \leq \log(N_e) < 4.8$.

S3. COMPOSITION EXTRACTION

For each N_e bin of interest, the relative composition of aforementioned primaries (H, He, N, Al and Fe) was extracted from the observed MMD, which contains EASs contributed by different primaries. The observed MMD was unfolded using Gold's algorithm [6] and response matrix ($\mathbf{R}_1$). The details on the $\mathbf{R}_1$, unfolding procedure and estimation of systematic uncertainty due to various sources are presented in the following subsections.

A. Response matrix

For each N_e bin, the MMD of each primary mass was normalized separately to have total probability equal to 1. Next the response matrix ($\mathbf{R}_1$) was generated using the normalized MMDs of the five primary masses. The element $R_{1,\alpha i}$ of the $\mathbf{R}_1$ represents the probability of an EAS initiated by the i^{th} simulated primary having the muon multiplicity value α for a given N_e and was calculated using the following steps,

1. Choose parameters from TABLE S1 for the i^{th} primary and calculate the μ and σ using Eq. S2 for shower size N_e .
2. Use Eq. S1 to get the probability of getting α muons,

$$R_{1,\alpha i} = NBD(\alpha; \mu, \sigma). \quad (\text{S3})$$

Hereafter, bin indices in reconstructed parameter space (muon multiplicity/shower size) are represented by Greek alphabets, and bin indices in primary parameter space (composition/energy) are represented by Latin alphabets. The top panel of FIG. S5 shows the response

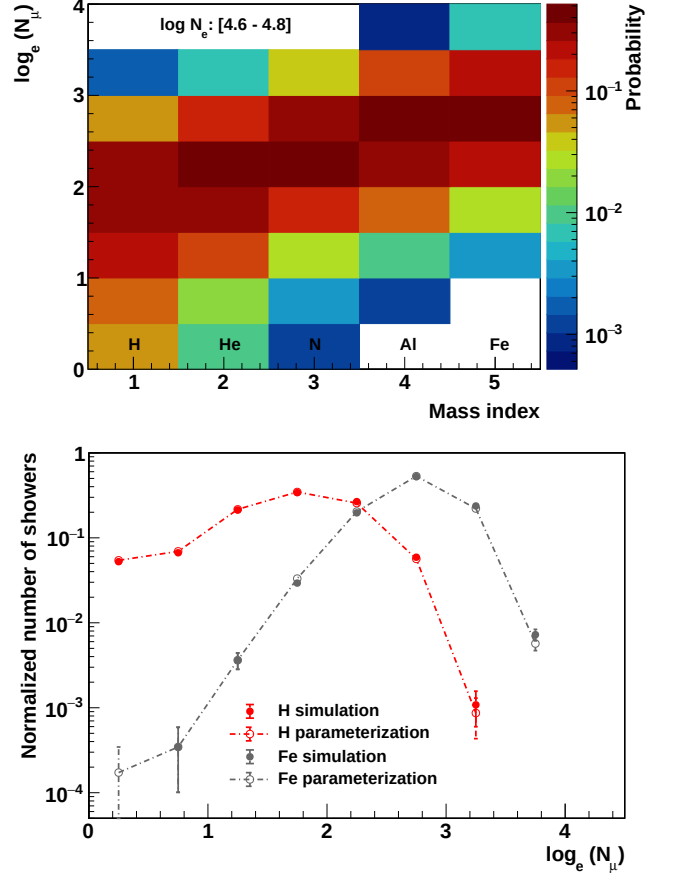


FIG. S5. Top: Response matrix ($\mathbf{R}_1$) for $4.6 \leq \log(N_e) < 4.8$. The color gradient represents the probability of an EAS initiated by a given simulated primary having the $\log_e(N_\mu)$ value. Bottom: Plot showing the comparison of MMDs on the natural log scale obtained from simulation (solid symbol) and parameterization (hollow symbol with a dashed connecting curve) for H and Fe primaries and the same shower size bin.

matrix ($\mathbf{R}_1$) for $4.6 \leq \log(N_e) < 4.8$, where the color gradient represents the probability values on the logarithmic scale. The x-axis represents the index (i) for different simulated primary groups used for this analysis (1, 2, 3, 4 and 5 represent H, He, N, Al and Fe, respectively), and the y-axis represents the value of N_μ on the natural logarithmic scale. In order to study the reliability of the $\mathbf{R}_1$, the dataset with spectral index -2.7 was split into two independent sub-datasets. The second sub-dataset was used for the parameterization and generating the $\mathbf{R}_1$ by following the steps mentioned above. The reliability of the $\mathbf{R}_1$ was tested by comparing the MMD for a given primary obtained from the first sub-dataset with the MMD generated for the same primary with the aid of random numbers using the corresponding column of $\mathbf{R}_1$ as a probability distribution, as shown in the bottom panel of FIG. S5 for H and Fe primaries and $4.6 \leq \log(N_e) < 4.8$. For parameterized MMDs, the same number of random events were generated as

in the simulated MMD. For both primaries, the MMDs obtained from simulation and parameterization show a good agreement with each other within statistical fluctuations.

B. Unfolding of observed MMD

Let $\vec{N}_\mu$ be the observed muon multiplicity vector, and $n(N_{\mu,\alpha})$ represents the number of EASs in the α^{th} bin of the observed MMD, and $\vec{A}$ be the composition vector such that $n(A_i)$ represents the number of EASs contributed by the i^{th} primary. The relative composition of i^{th} primary (a_i) is calculated as,

$$a_i = \frac{n(A_i)}{\sum_{j=1}^5 n(A_j)}. \quad (S4)$$

Gold's algorithm is an iterative method, and the i^{th} element of the composition vector ($\vec{A}^{k+1}$) for $(k+1)^{th}$ iteration was estimated from $\vec{A}^k$ as,

$$n(A_i^{k+1}) = n(A_i^k) \frac{(\mathbf{R}_1^T \mathbf{C}^T \mathbf{C} \vec{N}_\mu)_i}{\sum_{j=1}^5 (\mathbf{R}_1^T \mathbf{C}^T \mathbf{C} \mathbf{R}_1)_{ij} n(A_j^k)}, \quad (S5)$$

where $\mathbf{C}$ is the error matrix for the observed data such that $C_{\alpha\beta} = \delta_{\alpha\beta} / \sqrt{n(N_{\mu,\alpha})}$ (assuming Poisson) and $\delta_{\alpha\beta}$ is the Kronecker delta. The relative composition proposed by the GST model was used for the initial guess of $\vec{A}$. The average statistical uncertainty (σ) in the estimate of $\vec{A}$ increases, and the average bias (b) introduced by the unfolding algorithm decreases with an increase in the iteration number. The optimal balance between the statistical uncertainty and bias was achieved at the minimum value of weighted mean square error (WMSE) [7, 8]. The WMSE for k^{th} iteration is defined as,

$$WMSE^k = \frac{1}{m} \sum_{i=1}^m \frac{(\sigma_i^k)^2 + (b_i^k)^2}{A_i^k}, \quad (S6)$$

where $m = 5$ is the number of assumed primary groups. For further details on WMSE, see section 11.7 of [7]. At each iteration, σ and b were estimated by generating 50 samples from observed N_μ and the template of $\vec{A}$ ($\vec{A}^{temp}$) using the bootstrap method described in [9]. The $\vec{A}^{temp}$ represents the best guess of the $\vec{A}$ for the WMSE calculation. The iteration corresponding to minimum WMSE was identified as the optimal stopping iteration. The final relative composition was smoothed using a weighted running average (Savitzky–Golay filter) to reduce fluctuations in the observed N_e distribution subsample, which was used to measure energy distribution.

C. Proton relative composition

A comparison of the normalized observed MMD with the resultant curve obtained by adding the MMD curves

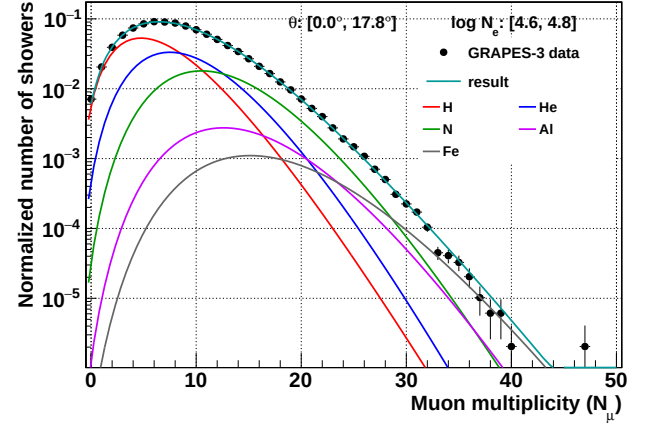


FIG. S6. Comparing the normalized observed MMD (black circles) with the resultant curve (cyan curve) obtained by adding the MMD curves of all simulated primaries (different colored curves) scaled by their relative composition estimated from the Gold's unfolding, for $4.6 \leq \log(N_e) < 4.8$.

TABLE S2. Relative composition of proton primary estimated using Gold's unfolding procedure for the GRAPES-3 data. The statistical and systematic uncertainties are mentioned.

N_e	$a_1 \pm \delta_1^{stat} + \delta_1^{sys} - \delta_1^{sys}$
1.26×10^4	$0.650 \pm 0.002 + 0.032 - 0.044$
2.00×10^4	$0.558 \pm 0.002 + 0.027 - 0.044$
3.16×10^4	$0.465 \pm 0.002 + 0.021 - 0.027$
5.01×10^4	$0.436 \pm 0.003 + 0.017 - 0.021$
7.94×10^4	$0.432 \pm 0.004 + 0.022 - 0.025$
1.26×10^5	$0.435 \pm 0.005 + 0.008 - 0.016$
2.00×10^5	$0.440 \pm 0.007 + 0.015 - 0.018$
3.16×10^5	$0.446 \pm 0.008 + 0.018 - 0.023$
5.01×10^5	$0.462 \pm 0.011 + 0.024 - 0.034$
7.94×10^5	$0.470 \pm 0.016 + 0.030 - 0.049$

of all simulated primaries scaled by their relative composition estimated from the unfolding procedure is shown in FIG. S6 for $4.6 \leq \log(N_e) < 4.8$. Both show a good agreement with a χ^2/dof of 3.0 (127.5/42).

The relative composition of proton primaries (a_1) for GRAPES-3 data is listed in TABLE S2 along with the statistical (δ_1^{stat}) and systematic (δ_1^{sys}) uncertainties.

D. Estimation of systematic uncertainties

The contribution from the following sources of systematic uncertainty was estimated.

(i) **Unfolding Algorithm:**— The composition of all primary groups was extracted using Gold's unfolding algorithm. The systematic uncertainty was calculated from the deviation in the relative composition of proton primaries obtained from the Bayesian unfolding [10] with Gold's unfolding for each N_e bin. The Bayesian unfold-

ing is also an iterative algorithm, and the i^{th} element of relative composition vector for $(k+1)^{th}$ iteration ($\vec{a}^{k+1}$) was estimated from the $\vec{a}^k$ as,

$$a_i^{k+1} = \frac{a_i^k}{\sum_{\alpha} R_{1,\alpha i}} \sum_{\alpha} \frac{R_{1,\alpha i} n_{\mu,\alpha}}{\sum_l R_{1,\alpha l} a_l^k}, \quad (S7)$$

where $\vec{n}_{\mu}$ is the normalized observed muon multiplicity vector, which is calculated as,

$$n_{\mu,\alpha} = \frac{n(N_{\mu,\alpha})}{\sum_{\beta} n(N_{\mu,\beta})}. \quad (S8)$$

The Bayesian unfolding was stopped at the same optimal stopping iteration as in the case of Gold's unfolding. The systematic uncertainty (δ_{UA}) for the proton primary is within 0% to 2%.

(ii) Initial guess/prior of $\vec{A}$ for unfolding:— The number of observed EASs weighted with the relative composition proposed by the GST model was used for the initial guess of the composition vector ($\vec{A}$). The convergence rate of the unfolding procedure depends on the choice of initial guess of $\vec{A}$. When the initial guess is closer to the expected solution, fewer number of iterations are required to obtain the estimate of $\vec{A}$. To calculate the systematic uncertainty due to the initial guess of $\vec{A}$, the unfolding procedure was repeated with initial guesses, where the total number of observed EASs are weighted according to the following relative compositions,

1. A uniform relative composition of 0.50, 0.3, 0.10, 0.04, and 0.06 for H, He, N, Al, and Fe, respectively.
2. Relative composition obtained from the H4a model.
3. A linearly varying relative composition obtained by connecting the GST model's relative composition in a shower size range of $10^{2.8}$ - $10^{6.2}$ for the corresponding primary.

The deviation in the relative composition of proton primaries obtained using initial guess 1, 2 or 3 from the reference composition was estimated. Three values of deviation were obtained for each N_e bin, and the maximum deviation among them on each side (positive and negative) represents the estimation of systematic uncertainty due to the initial guess of $\vec{A}$ (δ_P). It was found to be confined within $\pm 1\%$.

(iii) Bias from the unfolding procedure:— The regularized unfolding procedure adds a certain amount of bias (b) to control the amplification of the statistical fluctuations during the unfolding. For Gold's unfolding procedure, the number of iterations (k) acts as the regularization parameter. For each shower N_e of interest, the systematic bias (b_1) at the optimal stopping iteration for proton primary was calculated as,

$$b_1 = \frac{n(A_1) - n(A_1^{temp})}{n(A_1^{temp})}, \quad (S9)$$

where $n(A_1^{temp})$ is the number of EASs for the proton primary in the $\vec{A}^{temp}$. The systematic bias is within $+0.18\%/-0.15\%$.

(iv) Different spectral profiles to generate the response matrix:— The mean and standard deviation of MMD depend on the spectral profile of the energy spectrum of a given primary. For example, a flatter spectrum (smaller spectral slope) produces more weightage to higher muon multiplicity for a given N_e bin, hence, a slightly higher mean and vice versa. Therefore, using different spectra profiles to generate the response matrix produces a systematic uncertainty in estimating the relative composition. This uncertainty was calculated by repeating the unfolding procedure using response matrices that were generated assuming the spectral profile proposed by the GST and H4a models and calculating the deviation in the relative composition with respect to the relative composition obtained in this work (using spectrum with spectral index -2.7). The maximum deviation on either side (positive or negative) due to using both models represents the systematic uncertainty. The systematic uncertainty (δ_{SP}) is $+3.7\%/-5.8\%$ at shower size $10^{4.1}$ to $+0\%/-7.6\%$ at shower size $10^{5.9}$. For the proton primary, the difference between the spectral index proposed by the H4a and the one utilized in this analysis (-2.7) gradually changes from approximately $+0.04$ at shower size $10^{4.1}$ to approximately -0.20 at shower size $10^{5.9}$. This large difference at higher shower sizes is leading to a larger systematic deviation of -7.59% at shower size $10^{5.9}$.

(v) Smoothing algorithm:— The systematic uncertainty was calculated by smoothing the variation of the relative composition with the N_e using the 353HQ-twice smoothing algorithm [11] and calculating the deviation from the results of this work. The systematic uncertainty due to this effect (δ_S) was found to be within $+0.7\%$ and -4.1% .

(vi) Limited statistics of MC simulations:— The error in the fit parameters obtained during fitting MMDs of simulated primaries with Eq. S1 was used to calculate the systematic uncertainty. Assuming the fit parameters follow a Gaussian distribution where the value and error in the fit parameter represent the mean and standard deviation of the Gaussian distribution, 50 samples of response matrix were generated with the help of a Gaussian random number generator for each simulated primary and given shower size. The observed MMD was unfolded using these 50 samples of the response matrix using the same algorithm, initial guess of $\vec{A}$ and optimal stopping iteration to generate 50 samples of the $\vec{A}$. For a given shower size bin, the systematic uncertainty for the proton primary is represented by the standard deviation (σ_1) of 50 samples of the $\vec{A}$ and calculated as,

$$\sigma_1 = \sqrt{\frac{1}{50} \sum_{n=1}^{50} (n(A_{1,n}) - \langle n(A_1) \rangle)^2}, \quad (S10)$$

where $\langle n(A_1) \rangle$ is the mean number of showers in 50 sam-

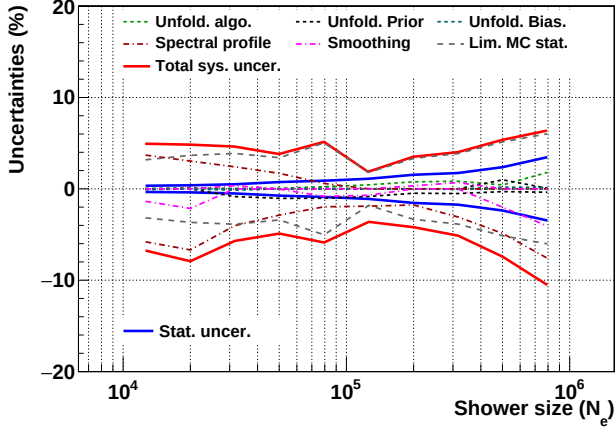


FIG. S7. Plot displaying the statistical (thick blue line) and total systematic uncertainty (thick red line) in the estimation of relative composition of proton primaries as a function of shower size. The total systematic uncertainty was calculated by adding the contribution of systematic uncertainty from different sources, represented by dashed lines with different colors, in quadrature.

ples of $\vec{A}$ for proton primaries and defined as,

$$\langle n(A_1) \rangle = \frac{1}{50} \sum_{n=1}^{50} n(A_{1,n}). \quad (\text{S11})$$

The value of systematic uncertainty first increases from 3.2% at $N_e = 10^{4.1}$ to 5.0% at $N_e = 10^{4.9}$ and then drops to 1.8% at $N_e = 10^{5.1}$. It further increases to 6.0% at $N_e = 10^{5.9}$. The drop in systematic uncertainty is due to the switching from reconstructed set-1 to set-2, as mentioned in section S1.

Total systematic (δ_{total}) uncertainty was calculated by adding the contribution of systematic uncertainty from different sources in quadrature and is shown in FIG. S7 along with the statistical and systematic uncertainty from different sources. The value of total systematic uncertainty, along with systematic uncertainty from different sources, is listed in the TABLE S3.

S4. PROTON ENERGY RESOLUTION

The energy resolution and bias (offset) were calculated with the aid of the MC simulations. As discussed in section S1, two simulated datasets (set-1 from 1 TeV to 10 PeV and set-2 above 100 TeV with a single spectrum of spectral index -2.7) were used to improve statistics for higher shower size. Each set was divided into two parts to have two independent datasets. The first sub-dataset of set-1 and set-2 was used to obtain a shower size to energy relation. For each shower size bin of interest, the median energy was plotted against the median shower size value in the same bin on a log-log scale, as represented by the violet circle symbol in the top panel of FIG. S8. The

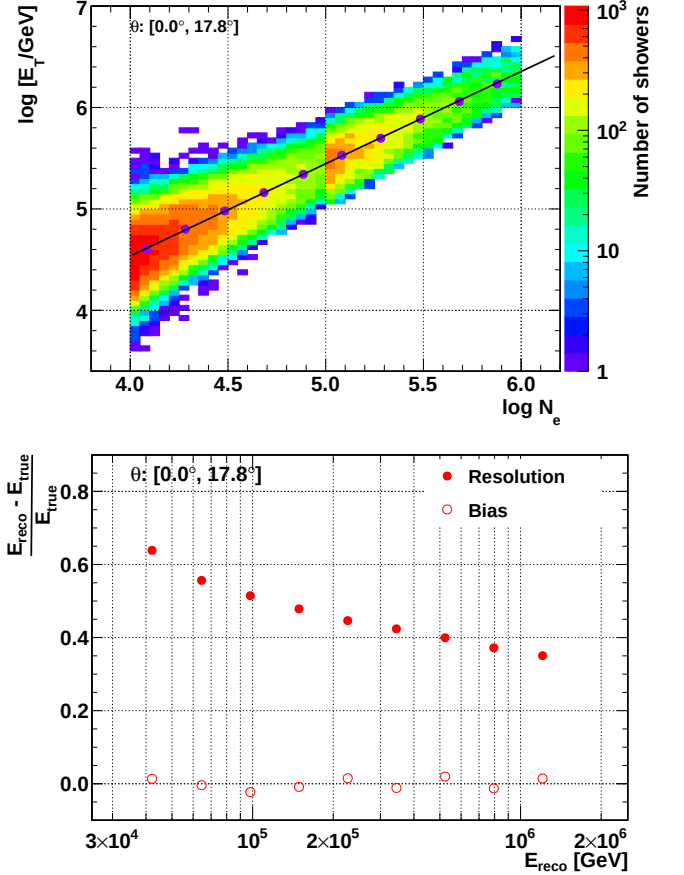


FIG. S8. Top: A two-dimensional representation of the shower size and primary energy distribution on the log-log scale, where color gradient represents the number of showers. The symbols represent the median energy corresponding to the median shower size with a bin width of 0.2 on a logarithmic scale, and the black line represents the linear fit. Bottom: The energy resolution and bias for the proton primaries as a function of reconstructed energy.

error bars are smaller than the size of the symbol. A two-dimensional distribution of shower size and energy is also displayed on a finer bin width, where the color gradient represents the number of EASs. The sharp change in the color gradient visible at $N_e = 10^{5.0}$ is due to the higher statistics of set-2. The shower size (N_e) to energy (E) relation can be well described with the simple form,

$$\log E = m \times \log N_e + c, \quad (\text{S12})$$

where c and m are the intercept and slope. The values of $c = 0.901 \pm 0.006$ and $m = 0.909 \pm 0.001$ were obtained from the fit represented by the black line in the top panel of FIG. S8.

The second sub-dataset for both set-1 and set-2 was used to calculate the energy resolution and bias. For each EAS, the reconstructed energy was estimated from shower size using Eq. S12. The distribution of relative deviation of reconstructed and true energy was generated for each shower size bin. The median and standard

TABLE S3. Systematic uncertainties (%) in estimating the relative composition of proton primary from various sources. Here systematic uncertainty δ_{UA} due to unfolding algorithm, δ_P due to initial guess/prior, b_1 due to bias from unfolding, δ_{SP} due to differential spectral profiles, δ_S due to smoothing algorithm, and σ_1 due to limited Monte Carlo statistics. Please see the text in [section S3D](#) for the description of each systematic uncertainty.

The last column represents the total systematic uncertainty, which was calculated by adding the contribution of systematic uncertainty from different sources in quadrature.

N_e	$\delta_{UA}(\%)$	$\delta_P(\%)$	$b_1(\%)$	$\delta_{SP}(\%)$	$\delta_S(\%)$	$\sigma_1(\%)$	$\delta_{total}(\%)$
1.26×10^4	+0.04	+0.00/-0.06	-0.07	+3.69/-5.80	-1.37	± 3.18	+4.87/-6.76
2.00×10^4	+0.01	+0.07/-0.00	-0.01	+3.04/-6.68	-2.15	± 3.67	+4.77/-7.92
3.16×10^4	+0.04	+0.00/-0.84	-0.15	+2.41/-4.03	+0.34	± 3.87	+4.57/-5.71
5.01×10^4	+0.04	+0.00/-1.04	+0.01	+1.70/-2.87	-0.04	± 3.41	+3.80/-4.90
7.94×10^4	+0.24	+0.00/-1.03	+0.08	+0.60/-1.97	-0.82	± 5.03	+5.07/-5.87
1.26×10^5	+0.44	+0.00/-0.83	+0.00	+0.00/-1.89	-0.70	± 1.81	+1.86/-3.61
2.00×10^5	+0.77	+0.00/-0.48	-0.02	+0.00/-1.76	+0.33	± 3.34	+3.44/-4.20
3.16×10^5	+0.84	+0.00/-0.51	-0.06	+0.00/-3.08	+0.69	± 3.86	+4.01/-5.12
5.01×10^5	+0.43	+0.99/-0.30	+0.18	+0.00/-4.88	-2.01	± 5.14	+5.25/-7.43
7.94×10^5	+1.80	+0.07/-0.37	+0.15	+0.00/-7.59	-4.07	± 6.03	+6.29/-10.52

deviation of the distribution represent the energy bias and resolution, respectively. The variation of the bias and resolution with reconstructed energy is shown in the bottom panel of [FIG. S8](#). The error bars for the energy bias are smaller than the symbol size. The energy bias is within $\pm 2\%$, and the energy resolution is $\sim 60\%$ at 50 TeV and improves to $\sim 35\%$ at 1.3 PeV. The GRAPES-3 experiment is located at an atmospheric depth (X) of around 800 g.cm^{-2} . The value of shower age (s) at shower size of $10^{4.1}$ (which corresponds to nearly 50 TeV energy) has been observed to be around 1.25. By using the relation given in Eq. 16.27 in Ref. [\[12\]](#), we obtain a shower maximum (X_{max}) = 560 g.cm^{-2} for 50 TeV and $s=1.25$. This calculation shows that the GRAPES-3 observational level is far below the shower maximum, which leads to relatively higher shower fluctuations. Consequently, GRAPES-3 has a higher value of energy resolution as compared to arrays such as HAWC, Tibet AS $_{\gamma}$, and LHAASO, which are located twice higher altitude and at about 200 g.cm^{-2} shallower depth, which is close to the shower maximum at 50 TeV.

S5. PROTON ENERGY SPECTRUM

A. Proton size distribution

For a given shower size bin, the relative composition of the proton primary was used as the statistical weight to obtain the corresponding shower size distribution from the observed distribution. Let the $n(N_{e,\alpha}^{obs})$ be the number of events in the α^{th} bin of observed shower size distribution, and $a_{1,\alpha}$ be the relative composition of the proton primaries in the same shower size bin. Then the number of EASs in the α^{th} bin of shower size distribution of the proton primaries is given as,

$$n(N_{e1,\alpha}) = n(N_{e,\alpha}^{obs}) a_{1,\alpha}. \quad (\text{S13})$$

The number of EASs observed by GRAPES-3 and the corresponding proton shower size distribution is listed in [TABLE S4](#) (with all selection cuts). The last column represents the statistical uncertainty in the estimation of proton events, which is calculated by propagating statistical uncertainties in the N_e^{obs} and a_1 for each shower size bin.

TABLE S4. The number of observed events by GRAPES-3 and corresponding proton events in different shower size bins. The statistical error in the estimation of proton events is listed in the last column.

N_e	$n(N_e^{obs})$	$n(N_{e1})$	$\delta N_{e1}(\%)$
1.26×10^4	3890720	2530350	0.34
2.00×10^4	1959300	1092590	0.39
3.16×10^4	970833	451097	0.51
5.01×10^4	486000	211886	0.76
7.94×10^4	246606	106595	0.91
1.26×10^5	126709	55074	1.14
2.00×10^5	65155	28664	1.59
3.16×10^5	33704	15043	1.81
5.01×10^5	17534	8097	2.49
7.94×10^5	8719	4100	3.63

B. Correction for miss and fake events

The occurrence of miss and fake events is primarily associated with the core resolution and angular resolution. Miss events refer to cases whose true cores are inside the fiducial area (50 m from the center of the array) and the true angle is less than 17.8° but the reconstructed core is either located outside the fiducial area or the reconstructed angle is greater than 17.8° . Conversely, fake events are characterized by true cores falling outside the fiducial area or true angles greater than 17.8° but the

reconstructed cores are inside the fiducial area and the reconstructed angle is less than 17.8° . Since the value of the core resolution improves with an increase in the shower size or energy, the fraction of the fake and miss event also decreases with an increase in the shower size or energy. It is shown in FIG. S10 that the core resolution first rapidly decreases with shower size up to nearly $N_e = 10^{5.0}$ and then begins to saturate. Therefore, saturation is observed in the fake events at the same shower size. The aforementioned shower size value corresponds to nearly 300 TeV energy. Thus, a similar saturation is observed in the miss events at a few hundred TeV. The correlations of the miss and fake event fractions with core resolution are 0.982 and 0.995, respectively. The fraction of miss and fake events is explored with the simulation because the information on the true and reconstructed cores is only available in the simulation.

The fraction of fake events (f_{fake}) for proton-initiated EASs as a function of shower size is shown in the top panel of FIG. S9. The correction was applied to the proton size distribution (N_{e1}) before the unfolding procedure by using,

$$n(\bar{N}_{e1,\alpha}) = n(N_{e1,\alpha})(1 - f_{fake,\alpha}), \quad (S14)$$

where $n(\bar{N}_{e1})$ is the number of EASs in the proton shower size distribution corrected for fake events, and α is the shower size bin index.

Similarly, the EAS belonging to energy distribution but not present in the shower size distribution is called a miss EAS. The bottom panel of FIG. S9 shows the fraction of the miss EAS (f_{miss}) as a function of primary energy for proton-initiated EAS. The correction was applied to the unfolded energy distribution of proton primaries ($\bar{E}$) as,

$$n(E_i) = \frac{n(\bar{E}_i)}{1 - f_{miss,i}}, \quad (S15)$$

where $n(\bar{E}_i)$ and $n(E_i)$ are the number of events in the i^{th} bin of unfolded energy distribution of proton primaries uncorrected and corrected for missed events, respectively. The statistical uncertainties in the fake and miss events were propagated to the final unfolded energy distribution.

C. Unfolding of proton energy distribution

The proton shower size distribution (N_{e1}), obtained in S4.A, was first corrected for the fake events using Eq. S14. The unfolded energy distribution for proton primaries ($\bar{E}$) was estimated from the corresponding shower size distribution corrected for fake events ($\bar{N}_{e1}$) using the same Gold's unfolding algorithm again, as used for the estimation of the relative composition. In this case, the response matrix ($\mathbf{R}_2$) was generated from the normalized shower size distribution for a given energy bin using the MC simulation dataset with a spectral slope of -2.7 . FIG. S11 shows the $\mathbf{R}_2$ for proton primaries in shower

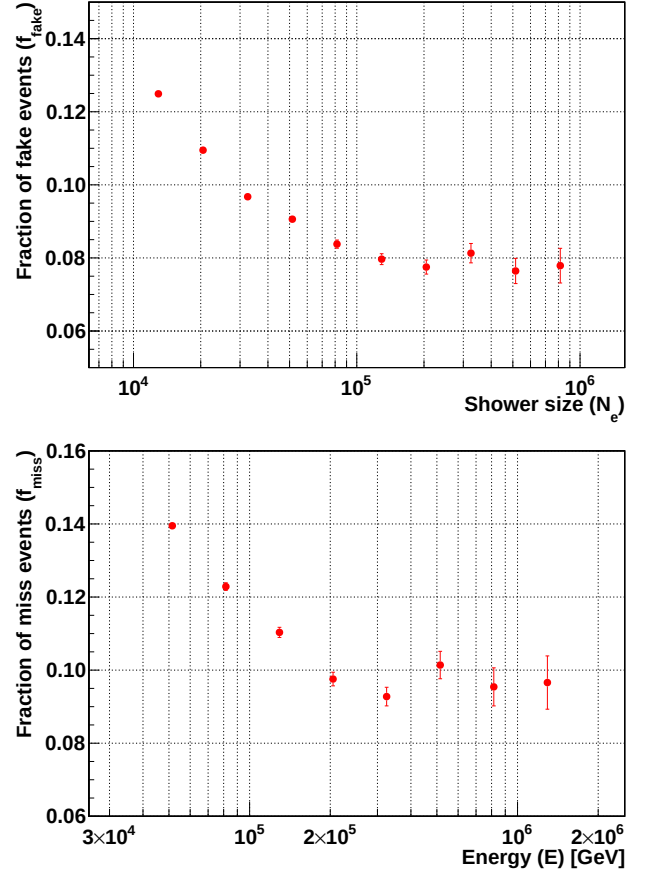


FIG. S9. Top: fraction of fake events (f_{fake}) as a function of shower size. Bottom: fraction of the miss events (f_{miss}) as a function of primary energy for proton-initiated EAS for the same dataset.

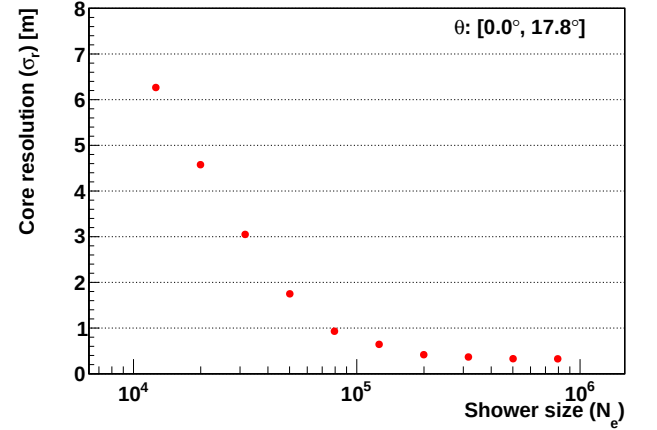


FIG. S10. Variation of the core resolution as a function of shower size.

size and energy phase space, such that $R_{2,\alpha i}$ represents the probability of a proton-initiated EAS belonging to i^{th} energy bin reconstructed in the α^{th} shower size bin. The initial guess for the energy distribution was selected

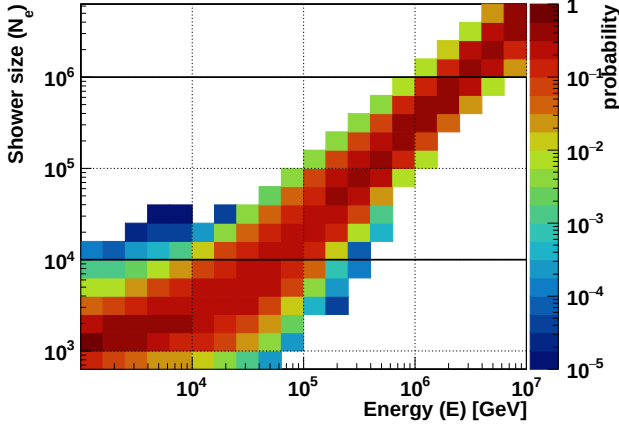


FIG. S11. The response matrix ($\mathbf{R}_2$) for proton primaries showing the probability (color gradient) of an EAS from energy bin E is reconstructed in shower size bin N_e . It is generated using the MC simulation dataset with a spectral slope of -2.7 . The two horizontal black lines enclose the shower size range of interest for this analysis.

with a spectral break near 200 TeV for fast convergence. By rewriting the Eq. S5 in terms of $\mathbf{R}_2$, $\vec{N}_{e1}$ and $\vec{E}$, the number of EASs in the i^{th} bin of the unfolded energy distribution for $(k+1)^{th}$ iteration ($\vec{E}^{k+1}$) was estimated iteratively as,

$$n(\vec{E}_i^{k+1}) = n(\vec{E}_i^k) \frac{(\mathbf{R}_2^T \mathbf{C}^T \mathbf{C} \vec{N}_{e1})_i}{\sum_j (\mathbf{R}_2^T \mathbf{C}^T \mathbf{C} \mathbf{R}_2)_{ij} n(\vec{E}_j^k)}, \quad (\text{S16})$$

where $\mathbf{C}$ is the error matrix for the $\vec{N}_{e1}$ such that $C_{\alpha\beta} = \delta_{\alpha\beta} / \sqrt{n(\vec{N}_{e1,\alpha})}$ (assuming Poisson) and $\mathbf{R}_2^T$ and $\mathbf{C}^T$ are the transpose of $\mathbf{R}_2$ and $\mathbf{C}$, respectively. Smoothing was applied after each iteration to control the statistical fluctuations. However, the final unfolded energy distribution was not smoothened. The same criterion of minimum WMSE was used to stop the unfolding iterations. Finally, the $\vec{E}$ was corrected for the miss events using Eq. S15.

D. Unfolded proton energy spectrum

The unfolded number of events for each energy bin of the energy distribution of proton primaries is listed in the second column, and the value of the flux ($\Phi(E)$) is listed in the third column of TABLE S5 along with the corresponding statistical and systematic uncertainties. The energy value in the first column is the geometric mid-value of the energy bin on the log scale. The statistical uncertainty is 0.4% at 50.1 TeV and increases to 3.7% at 1.3 PeV. On the other hand, the total systematic uncertainty is +4.5%/-6.9% at 50.1 TeV and increases to +11.0%/-12.9% at 1.3 PeV.

TABLE S5. Summary of the unfolded number of events in the energy distribution of proton primaries for GRAPES-3 data estimated using Gold's unfolding algorithm. The values of differential flux ($\Phi(E)$) and the corresponding statistical and total systematic uncertainties are listed in the third column.

Energy [GeV]	Number of events	$\Phi \pm \delta\Phi_{stat.} + \delta\Phi_{sys.} - \delta\Phi_{sys.}$ [$m^{-2} sr^{-1} s^{-1} GeV^{-1}$]
5.01×10^4	2797488	$(2.80 \pm 0.01 + 0.13 - 0.19) \times 10^{-9}$
7.94×10^4	1139870	$(6.67 \pm 0.03 + 0.36 - 0.53) \times 10^{-10}$
1.26×10^5	450551	$(1.63 \pm 0.01 + 0.09 - 0.11) \times 10^{-10}$
2.00×10^5	189363	$(4.30 \pm 0.05 + 0.32 - 0.41) \times 10^{-11}$
3.16×10^5	89306	$(1.28 \pm 0.02 + 0.11 - 0.12) \times 10^{-11}$
5.01×10^5	44715	$(4.04 \pm 0.08 + 0.27 - 0.36) \times 10^{-12}$
7.94×10^5	21194	$(1.21 \pm 0.03 + 0.08 - 0.11) \times 10^{-12}$
1.26×10^6	9928	$(3.57 \pm 0.13 + 0.39 - 0.46) \times 10^{-13}$

E. Estimation of systematic uncertainties

The contribution from the following sources of systematic uncertainty in the estimation of the energy spectrum of proton primaries using Gold's unfolding algorithm was estimated.

(i) **Unfolding algorithm:**— The unfolding of proton shower size distribution into the energy distribution was repeated with the Bayesian unfolding algorithm using the same initial guess and optimal stopping iteration as in the case of Gold's algorithm. The systematic uncertainty was calculated from the deviation in the flux of proton primaries obtained from the Bayesian unfolding with Gold's unfolding. The systematic uncertainty (δ'_{UA}) is within -0.02% to $+0.11\%$.

(ii) **Initial guess/prior of $\vec{E}$ for unfolding:**— The initial guess for energy distribution of proton primaries $\vec{E}$ was selected closer to the expected energy distribution for faster convergence. The unfolding procedure was repeated with the following initial guesses,

1. Proton energy distribution with spectral profile proposed by the GST model.
2. Proton energy distribution with spectral profile proposed by the H4a model.
3. An energy distribution with a differential spectral index of -2.7 .

The systematic uncertainty (δ'_P) was calculated from the maximum deviation (positive and negative) among the proton flux obtained using initial guess 1, 2 or 3 with the proton flux estimated in this work. It is found to be within -0.73% to $+1.20\%$.

(iii) **Bias from the unfolding procedure:**— The systematic bias in i^{th} energy bin b'_i at the optimal stopping iteration was calculated as,

$$b'_i = \frac{n(E_i) - n(E_i^{temp})}{n(E_i^{temp})}, \quad (\text{S17})$$

where $n(E)$ and $n(E^{temp})$ are the number of showers in the energy distribution and the corresponding template distribution for proton primaries. The systematic bias is within $\pm 0.40\%$.

(iv) GRAPES-3 acceptance:— The GRAPES-3 acceptance was modeled with the Richards function, and the value and error in fit parameters were used to calculate the systematic uncertainty. The 1000 values of acceptance were generated for each energy bin using a Gaussian random number generator, taking the value and error of fit parameters as the mean and standard deviation of the Gaussian function. The standard deviation of 1000 values of the acceptance in a given energy bin represents the uncertainty in the same energy bin. The uncertainty due to the statistical fluctuations in the acceptance is $\pm 2.05\%$ at 50.1 TeV and $\pm 0.04\%$ at 1.3 PeV. The systematic uncertainty associated with the selection of the fiducial area was also calculated. The entire procedure was repeated with a fiducial area of radius 40 m from the center of the array (instead of a radius of 50 m that is used in the main analysis). The proton energy spectrum shows a systematic deviation (δ_{Acc}) within 0.0% to $+3.9\%$.

(v) Different spectral profiles to generate the response matrix:— The spectral profiles proposed by the GST composition model and a differential spectral slope of -2.5 were used alternatively to generate the response matrices. The systematic uncertainty (δ'_{SP}) was calculated from the maximum deviation (positive or negative) in the proton flux obtained using the alternate response matrices with respect to the proton flux obtained in this work and was found to be within $+0.71\%$ to -4.15% .

(vi) Limited statistics of MC simulations:— In order to calculate the systematic error due to limited statistics, 50 samples of the response matrix were generated with the help of a random number generator using the response matrix used in this analysis as the probability distribution. The unfolding procedure was repeated with these samples to obtain 50 samples of unfolded energy distribution ($\vec{E}$). The systematic uncertainty in a given energy bin i is represented by the standard deviation (σ'_i) of 50 samples of the $\vec{E}$ and calculated as,

$$\sigma'_i = \sqrt{\frac{1}{50} \sum_{n=1}^{50} (n(E_{i,n}) - \langle n(E_i) \rangle)^2}, \quad (S18)$$

where $\langle n(E_i) \rangle$ is the average number of showers in the i^{th} bin of 50 samples of $\vec{E}$ for proton primaries and defined as,

$$\langle n(E_i) \rangle = \frac{1}{50} \sum_{n=1}^{50} E_{i,n}. \quad (S19)$$

The value of systematic uncertainty increases from 0.42% at 50.1 TeV to 2.06% at 1.3 PeV.

(vii) Systematic uncertainty in composition:—

The systematic uncertainty in the estimation of the relative composition of proton primary (see section S3 D) was propagated into the unfolded proton energy spectrum. The iterative Gold's unfolding algorithm was used, and the optimal stopping iteration (k) was identified by the minimum WMSE. From Gold's algorithm, the number of showers in the i^{th} bin of energy distribution of proton primaries at k^{th} iteration (E^k) is calculate using Eq. S16. Using the expression $C_{\alpha\beta} = \delta_{\alpha\beta} / \sqrt{n(\bar{N}_{e1,\alpha})}$, the Eq. S16 can be simplified as,

$$n(\bar{E}_i^k) = \frac{n(\bar{E}_i^{k-1}) \sum_{\alpha} R_{2,\alpha i}}{\sum_j R_{2,ij}^{mod} n(\bar{E}_j^{k-1})}, \quad (S20)$$

$$R_{2,ij}^{mod} = \sum_{\alpha} \frac{R_{2,\alpha i} R_{2,\alpha j}}{n(\bar{N}_{e1,\alpha})}.$$

The systematic uncertainty for each energy bin is calculated by propagating the statistical uncertainty in $\vec{a}_1$ (δa_1) iteratively to i^{th} bin of the energy distribution ($\vec{E}$) obtaining at the optimal stopping iteration (k) as,

$$\frac{\partial n(\bar{E}_i^k)}{\partial a_{1,\alpha}} = n(\bar{E}_i^k) \left[t1 + \frac{1.0}{\sum_j R_{2,ij}^{mod} n(\bar{E}_j^{k-1})} (t2 - t3) \right],$$

$$t1 = \frac{1.0}{n(\bar{E}_i^{k-1})} \frac{\partial n(\bar{E}_i^{k-1})}{\partial a_{1,\alpha}},$$

$$t2 = \sum_j \frac{R_{2,\alpha j} R_{2,\alpha i}}{a_{1,\alpha} n(\bar{N}_{e1,\alpha})} n(\bar{E}_j^{k-1}),$$

$$t3 = \sum_j R_{2,ji}^{mod} \frac{\partial n(\bar{E}_j^{k-1})}{\partial a_{1,\alpha}}, \quad (S21)$$

$$\delta \bar{E}_i^k = \sqrt{\sum_{\alpha} \delta a_{1,\alpha}^2 \left(\frac{\partial n(\bar{E}_i^k)}{\partial a_{1,\alpha}} \right)^2}, \quad (S22)$$

with $\partial n(\bar{E}_i^0) / \partial a_{1,\alpha} = 0$. Finally, the systematic error in i^{th} bin of the proton energy distribution corrected for miss events (δE_i) is calculated as,

$$\delta E_i = \frac{\delta \bar{E}_i^k}{1 - f_{miss,i}}. \quad (S23)$$

The systematic uncertainty is $+3.97\%/-6.49\%$ at 50.1 TeV and $+10.75\%/-12.37\%$ at 1.3 PeV.

Total systematic uncertainty (δ'_{total}) was calculated by adding the contribution of systematic uncertainty from different sources in quadrature. The value of total systematic uncertainty and systematic uncertainty from different sources is listed in the TABLE S6. The δ_{Acc} represents the combined (adding in quadrature) systematic uncertainty due to statistical fluctuations in the GRAPES-3 acceptance and selection of the fiducial area.

TABLE S6. Systematic uncertainties (%) in estimating the energy spectrum of proton primaries using Gold's unfolding algorithm. Here the systematic uncertainty δ'_{UA} due to unfolding algorithm, δ'_P due to initial guess/prior, b' due to bias from unfolding, δ_{Acc} (%) due to acceptance, δ'_{SP} due to differential spectral profile, σ' due to limited Monte Carlo statistics, and δE (%) due to composition. Please see the text in [section S5 E](#) for the description of each systematic uncertainty. The last column represents the total systematic uncertainty, which was calculated by adding the contribution of systematic uncertainty from different sources in quadrature.

Energy [GeV]	δ'_{UA} (%)	δ'_P (%)	b' (%)	δ_{Acc} (%)	δ'_{SP} (%)	σ' (%)	δE (%)	δ'_{total} (%)
5.01×10^4	-0.02	+0.12/-0.73	+0.37	+2.08/-2.05	+0.10/-0.90	± 0.42	+3.97/-6.49	+4.52/-6.92
7.94×10^4	+0.07	+0.54/-0.00	+0.00	+1.66/-0.71	+0.31/-1.73	± 0.65	+5.02/-7.71	+5.36/-7.96
1.26×10^5	+0.09	+1.20/-0.07	-0.33	+2.46/-0.22	+0.71/-1.96	± 0.85	+4.98/-6.52	+5.79/-6.88
2.00×10^5	+0.08	+0.60/-0.09	-0.31	+3.49/-0.07	+0.66/-1.33	± 0.94	+6.47/-9.48	+7.47/-9.63
3.16×10^5	+0.09	+0.00/-0.49	+0.01	+3.91/-0.04	+0.00/-0.69	± 1.02	+7.58/-9.33	+8.60/-9.43
5.01×10^5	+0.11	+0.03/-0.71	+0.22	+2.97/-0.04	+0.07/-2.50	± 0.96	+5.93/-8.36	+6.71/-8.80
7.94×10^5	+0.08	+0.05/-0.11	+0.15	+1.65/-0.04	+0.00/-4.15	± 1.47	+5.86/-7.51	+6.27/-8.70
1.26×10^6	+0.08	+0.16/-0.00	+0.05	+0.84/-0.04	+0.00/-2.97	± 2.06	+10.75/-12.37	+10.98/-12.89

S6. CALCULATION OF SIGNIFICANCE

The p-value was calculated as,

$$p = \int_{\Delta\chi^2}^{\infty} f(x; 3) dx, \quad (S24)$$

where $f(x; 3)$ is the χ^2 distribution with three degrees of freedom. The p-value is 1.35×10^{-193} . By following [13], the corresponding significance (s) was calculated as,

$$s = \phi^{-1}(1 - p), \quad (S25)$$

where ϕ^{-1} is the inverse of the cumulative distribution of the standard Gaussian. An improvement in the fitting ($\Delta\chi^2$) of 894.54 for a reduction of 3 degrees of freedom favors the SBPL function with a significance of $\sim 29.7\sigma$ over the PL function.

S7. GRAPES-3 DATA SUMMARY

For this analysis, the data recorded by the EAS array as well as G3MT from 1 January 2014 to 26 October 2015 was used, which comprises a total of 1.75×10^9 EAS events. The number of EASs surviving after each quality cut and their percentage are listed in the second and third columns of [TABLE S7](#), respectively. About 10% of triggered events were rejected due to the imposition of a cut related to more than $\pm 2\%$ deviation of the daily shower size spectrum compared with the average monthly shower size spectrum or showed a systematic increase/decrease trend. The problem is mainly related to the abnormalities in the associated electronic channels, such as analog-to-digital converters or time-to-digital converters or data-acquisition electronics in the scintillator detector system. Nearly 23% events were rejected due to some clock problems when the event timing information recorded by the scintillator array could not be matched with the muon data recorded by the muon

telescope. The fiducial area selection within a 50 m radius from the center of the array retained nearly 50% events. Next, the imposition of a zenith angle selection to be below 17.8° retained about one-third of the surviving events with respect to the previous cuts. Further, imposing a selection on the EAS cores to be beyond a 60 m radius from the center of the muon telescope to reduce the hadron-punch through effect to be less than 2% further reduced the data sample by nearly 50%. Finally, the selection of the EASs with shower size between $10^{4.0}$ - $10^{6.0}$ reduced the surviving data sample to nearly 11%. After applying all the quality cuts, the final number of surviving EASs was 7.81×10^6 , representing 0.4% of the triggered EASs.

S8. THE DIFFERENCE BETWEEN PREVIOUS AND PRESENT ANALYSIS

The current result shows a spectral hardening of about 165 TeV, but not the previous results [14, 15], is primarily attributed to the implementation of the unfolding method, which is more sophisticated and reliable as compared to the methodology employed in the previous analysis. Different post-LHC hadronic interaction models yield differences in the muon number, and the difference among these models is seen to be small [16]. However, the observed spectral hardening is primarily related to the analysis procedure, as explained below.

For an assumed spectral profile, EAS simulation dataset was generated for proton primary. The reconstructed shower size (N_e) of the simulated events was binned according to a bin width of $10^{0.2}$ over $N_e = 10^{4.0}$ - $10^{6.0}$. Since the energy of each EAS event is known, the energy distribution for the N_e bins were obtained separately. Each energy distribution was approximated with a Gaussian function. The mean and standard deviation of this Gaussian function were used to convert the observed N_e s into an energy distribution with the

TABLE S7. The tabulation of the number of observed EASs after passing each quality cut. The quality cuts are listed in the first column, and the number of surviving EASs after applying the quality cut is listed in the second column. The last column represents the percentage of surviving EASs after each quality cut.

Quality cut	Number of surviving EASs	% of surviving EASs
1. Triggered	1.75×10^9	100.0
2. Abnormal days based on size spectrum	1.58×10^9	90.0
3. Successful event matching & muon tracking	1.17×10^9	66.8
4. Angle and NKG reconstruction	8.47×10^8	48.3
5. Shower age (s) between 0.2 and 1.8	8.41×10^8	48.0
6. Fiducial area within 50 m radius	3.96×10^8	22.6
7. Zenith angle $< 17.8^\circ$	1.33×10^8	7.5
8. Hadron punch-through $< 2\%$	6.27×10^7	3.6
9. Shower size (N_e) between $10^{4.0}$ and $10^{6.0}$	7.81×10^6	0.4

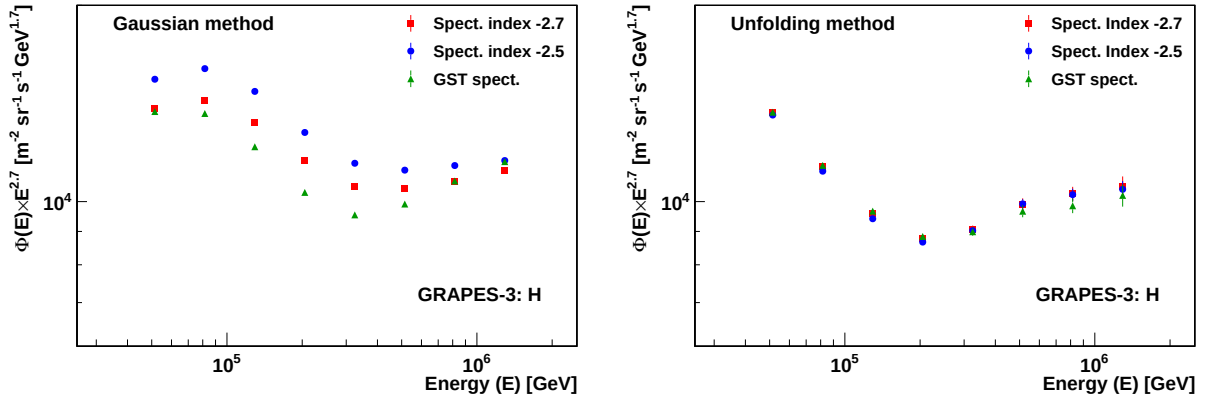


FIG. S12. Left: A comparison of the proton energy spectrum obtained for the GRAPES-3 data by employing the Gaussian method. Energy probability distributions are generated from three datasets with spectral profiles, including a spectral index of -2.5 , a spectral index of -2.7 , and the profile suggested by the GST model. Right: A comparison of the proton energy spectrum obtained for the GRAPES-3 data by employing the unfolding method. The response matrices are obtained with the same spectral profiles as mentioned above.

help of a Gaussian random number generator. All 10 energy distributions were combined to yield the final energy distribution for the observed data for the proton primary. We refer to this as the Gaussian method. From the energy distribution, the proton energy spectrum was obtained following the procedure discussed in the main paper. With three different assumed spectral profiles to generate the simulation dataset, such as one with a spectral index of -2.5 , a spectral index of -2.7 , and the profile suggested by the GST model and following the Gaussian method as discussed above, the energy spectra were obtained which are shown in the left panel of FIG. S12. A systematic deviation in the flux within 0% to 23% is observed between the results obtained by using the spectral profile proposed by the GST model and a spectral index of -2.5 . Apart from different flux values, the shape of the energy spectra is different. A spectral break is observed in each case, however the position of it varies from about 300 TeV to 500 TeV for the GST model to a spectral index of -2.5 . The right panel of FIG. S12 compares the

results obtained by employing the unfolding method with the response matrices generated with simulation datasets with the three above-mentioned spectral profiles. It exhibits a systematic deviation within $+0.71\%$ and -4.15% among the results obtained in the case of unfolding. This shows that the results are dependent on the chosen spectral profile for the simulation in the case of the Gaussian method, whereas they are almost independent in the case of the unfolding method.

To test the reliability of these methods, we carried out a Monte Carlo (MC) test where the input spectrum is known to compare. The MC dataset was generated using the proton energy spectrum as proposed by the GST model. The dataset was divided into two parts where one part served as the reference spectrum and the second part was used to test the different spectral profiles. We explored three different spectral profiles, one with a spectral index of -2.5 , one with a spectral index of -2.7 , and the GST model profile. For each profile, the energy distribution was generated using the Gaussian method,

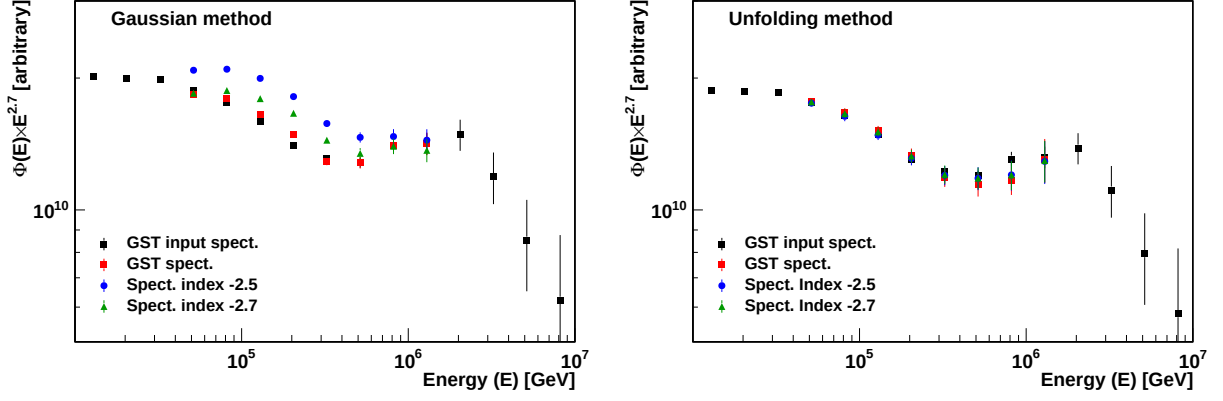


FIG. S13. Left: A comparison of the input proton energy spectrum proposed by the GST composition model with the spectra obtained by employing the Gaussian method. Energy probability distributions are generated from three datasets with spectral profiles, including a spectral index of -2.5 , a spectral index of -2.7 , and the profile suggested by the GST model. Right: A comparison of the input proton energy spectrum proposed by the GST composition model with the spectra obtained by employing the unfolding method. The response matrices are obtained with the same spectral profiles as mentioned above.

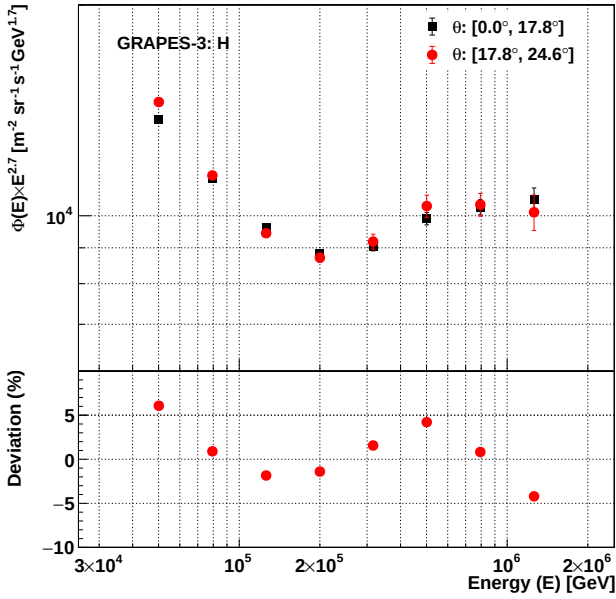


FIG. S14. Top: A comparison of the measured proton energy spectra obtained for two adjacent zenith angle bins (0.0° - 17.8° and 17.8° - 24.6°). Bottom: Difference of the proton energy spectrum obtained for the second zenith angle bin (17.8° - 24.6°) with respect to the first one.

as discussed above. In the left panel of FIG. S13, the energy spectra obtained in these three cases are compared with the reference spectrum. It is observed that the reference spectrum is accurately reproduced when the energy distributions were obtained from the spectrum with the spectral profile same as the reference spectrum (GST model in this example). However, the output spectra ex-

hibit systematic deviations from the reference spectrum in other cases. In this exercise, a systematic deviation within 2% to 22% is observed between the results obtained by using the spectral profile proposed by the GST model and a spectral index of -2.5 . In the case of unfolding, the response matrices were generated using the three above-mentioned spectral profiles. In this case, all three spectral profiles have accurately reproduced the reference spectrum, as shown in the right panel of FIG. S13. The observed systematic deviation among the results obtained in this case is within -2% to $+3\%$.

S9. PROTON ENERGY SPECTRUM WITH THE NEXT ZENITH ANGLE BIN

In this analysis, nearly vertical showers with a zenith angle (θ) range of 0 to 17.8° or $\sec(\theta) = 0$ to 1.05 were selected to reduce the effect of the shower attenuation caused due to the larger path traveled by the EAS particles in the atmosphere. By assuming the isotropic distribution of the cosmic rays, the observed energy spectra for different zenith angle ranges are expected to be the same for each zenith angle bin. However, due to the larger attenuation experienced by more inclined showers, the shower-to-shower fluctuations are more for smaller shower size at the observational level for a given energy. Thus, it leads to higher systematic uncertainties.

To study the effect of the zenith angle on the final result, the entire process is repeated for the next zenith angle bin ($\sec(\theta) = 1.05$ to 1.1 or $\theta = 17.8^\circ$ to 24.6°). The top panel of FIG. S14 displays a comparison between the proton energy spectra obtained for the two zenith angle bins. The difference of the proton energy spectrum obtained for the second zenith angle bin with respect to the first one is observed to be within $\pm 5\%$.

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